

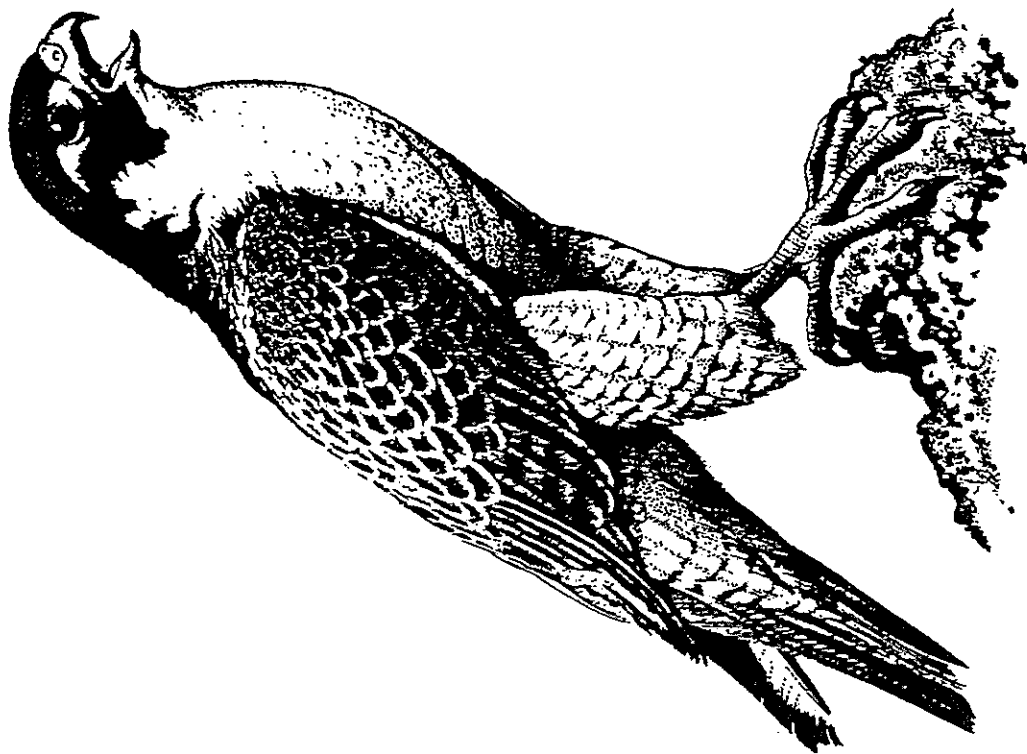
THE PEREGRINE FALCON

Peregrine falcons are found all over the world. In Australia, these birds have no natural predators, yet their numbers have declined alarmingly since 1950. Studies have shown that the number of eggs laid by the parent birds are about the same from year to year. However, the eggshell has gradually become thinner since the 1950s. The thickness of some eggshells has decreased by 20 per cent. It is believed that the thinner-shelled eggs break more easily in the nest, so that far fewer young hatch.

A number of causes may have produced the shell thinning, e.g. climate and disease. However, the changes in eggshell thickness coincide with the massive use of the insecticide DDT in the 1950s and 1960s. The thinnest shells were found in areas where the most DDT had been sprayed. Most investigators believe that DDT is responsible for the thinning of the eggshells. Insect-eating birds might be killed through the build up of insecticide in their bodies. Seed-eating birds may die from the same cause.

But the peregrine falcon is a flesh-eater. It is a hunter, preying on smaller animals.

HOW COULD DDT REACH A BIRD LIKE THE FALCON?



INSECTICIDE LEVELS IN ANIMALS

Some insecticides are
non-biodegradable.

What is meant by this?



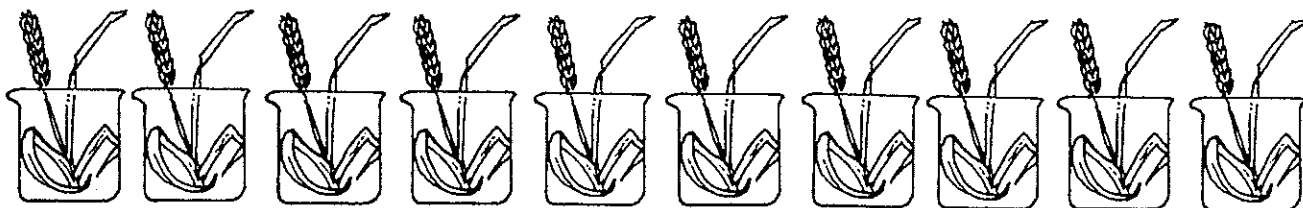
Arctic polar bears and seals in the Antarctic have been found to contain insecticides. How could the pesticide get into these animals when they have had no *direct* contact with it?

This demonstration should make it clear.

Apparatus in the laboratory will be used to represent real animals and plants.

There will be

10 WHEAT PLANTS



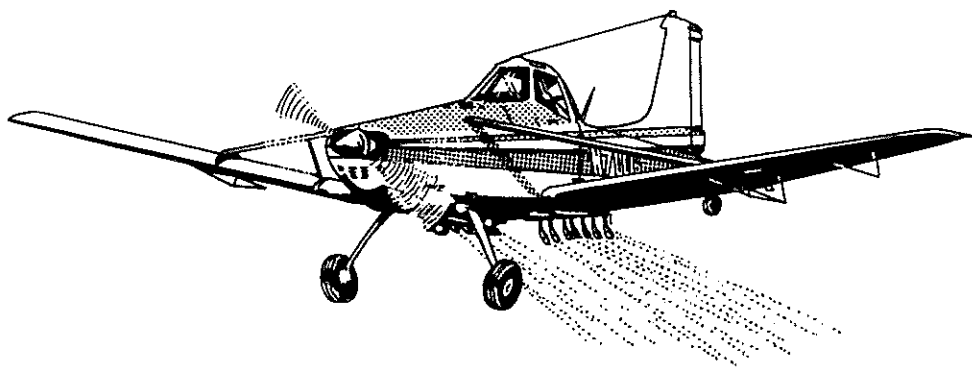
3 GRASSHOPPERS



and 1 KOOKABURRA



Suppose a wheatcrop is sprayed with an insecticide.



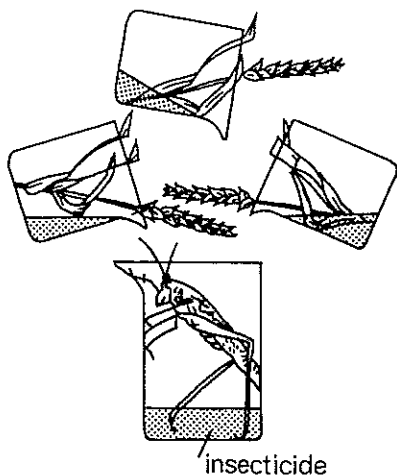
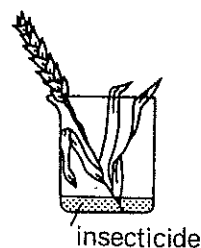
A little insecticide remains in each wheat plant.

1. What is the height of the wheat plant beaker? _____ cm.
2. What height is taken up by the insecticide (sand)? _____ cm.

Make a fraction from 1 and 2

That is, $\frac{2}{1} =$ _____

Each grasshopper then eats 3 wheat plants.



3. What height is each grasshopper?
_____ centimetres.
4. What height is taken up by the
insecticide?
_____ centimetres.

Where has this insecticide in the
grasshoppers come from?

A kookaburra now eats those 3 grasshoppers.

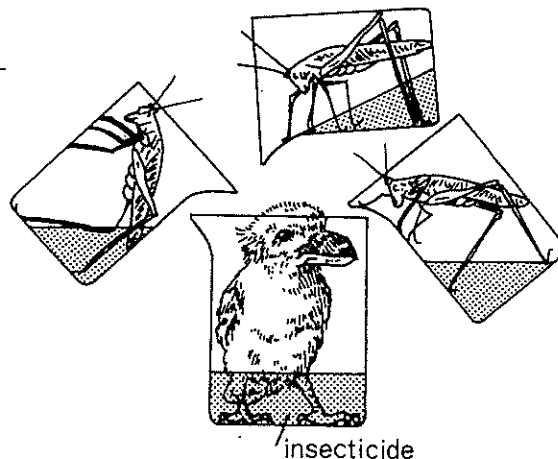
5. What height is the kookaburra?
_____ centimetres.

6. What height is taken up by the
insecticide? _____
centimetres.

Make a fraction from (5) and (6)

$$\frac{(6)}{(5)} = \underline{\hspace{2cm}}$$

What do you notice about the two
fractions?



Which organism has more insecticide in its body? _____

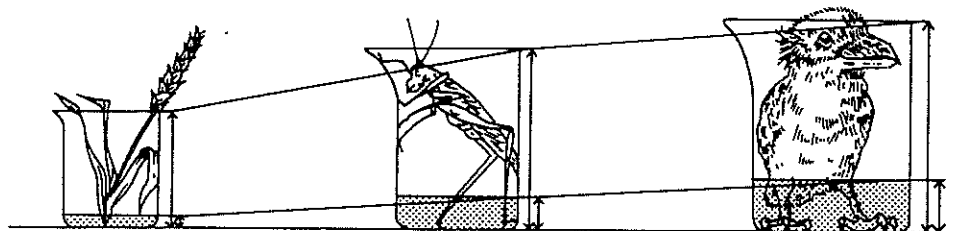
This is *exactly* what happens to some insecticides in the bodies of animals.

WHY IS THIS SO?

Particular insecticides such as DDT, called organochloride insecticides,

- *Are not got rid of from the body as wastes.*
- *Are not broken down into simple harmless substances in the body.*
- *Can increase in concentration inside an animal's body (stored in body fat) over a period of time.*

Here is the FOOD CHAIN you have demonstrated in this activity.



Write down a food chain you can expect in your garden. It should be on where
the insecticide level could build up in animals high in the chain.

Explain how innocent animals (e.g. ladybirds, honeyeaters) can be harmed by insecticides.

What effect could this have on harmful insects pests in a garden?

Explain how we could be harmed by insecticides in the foods we eat. (Refer to a food chain.)

Now explain how the Arctic polar bears and Antarctic seals could have picked up traces of insecticides.

Even human milk has been found to contain DDT. Suggest a food chain which might result in this situation.

BRAIN TEASER

How useful was the beaker model for showing DDT residues?